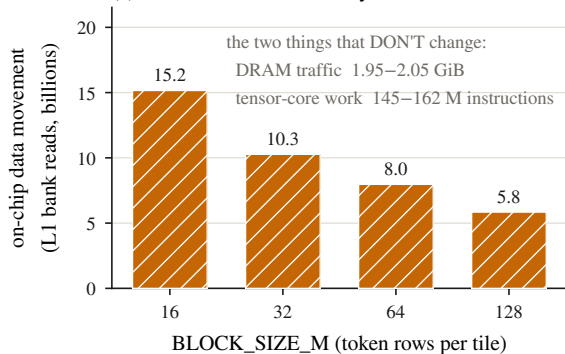


A100-SXM4-40GB | Qwen3-30B-A3B MoE layer, 7744 prefill tokens, real ShareGPT routing. (a) ncu, whole layer, both grouped GEMMs. (b) the 16 least-loaded experts (1821 real tokens, 3–217 each) run alone on 54 SMs, each tile size at its own energy-optimal clock, 3 passes, mean $\pm$ sd.

A 16-row tile reuses each B fragment one eighth as often as a 128-row tile, so SM throughput falls from 79% of peak to 24% while DRAM traffic and tensor-core work stay put. Padding is real -- 76% of the light partition's rows at BLOCK_SIZE_M=128 -- and it is not what this kernel is short of.

(a) what a smaller tile actually costs



(b) and what it costs in energy

